

Monitoring and Understanding Notebook-Based Learning with Jupyter

A Two-Tier Dashboard for Students and Instructors

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As programming becomes increasingly integrated into scientific and educational workflows, the need to understand how users interact with computational notebooks has grown. This project introduces a JupyterLab dashboard extension that monitors and visualizes notebook activity, including notebook openings, cell execution outcomes, errors, and retries. The system produces summary statistics and visual timelines that portray how the workflow unfolded. Two complementary versions are proposed: a student version, focused on self-assessment and awareness of their own coding patterns, and a teacher version, designed to reveal common difficulties across learners. To evaluate the effectiveness and usability of these tools, surveys will be conducted with both students and instructors. Together, these approaches provide a richer understanding of problem-solving behaviors within interactive coding environments and evidence of the dashboard's practical value.

CCS Concepts: • **Applied computing** → **Interactive learning environments**; • **Human-centered computing** → **Empirical studies in visualization**.

Additional Key Words and Phrases: Jupyter Notebooks, Learning Analytics, Educational Dashboards, Student Self-Assessment, Instructor Insights, Interactive Coding Environments

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1 INTRODUCTION

Computational notebooks such as Jupyter have become central tools in scientific computing, teaching, and data analysis. Their interactive structure—mixing narrative text, code, and visual output—makes them highly suitable for inquiry-based learning in programming education [5]. Yet despite their pedagogical advantages, instructors often lack visibility into how students actually work within these environments. Students, on the other hand, often have little opportunity to reflect on their own workflow, error patterns, or problem-solving behavior. Research in programming education consistently shows that learners repeat similar types of mistakes, struggle to identify the sources of their errors, and often adopt inefficient trial-and-error strategies. At the same time, teachers typically only see final code submissions, which provide limited insight into the learners' process [6]. This creates a gap between the correctness of the final code and the path the student took to reach it. Understanding this process is crucial to improve teaching interventions and help students develop their metacognitive skills.

Learning dashboards have been shown to enhance students' self-awareness, foster reflection, and offer instructors actionable insights into learner behavior. However, the majority of existing dashboards focus on high-level learning management system data rather than fine grained programming behavior within computational notebooks [4].

This project addresses this gap by designing a JupyterLab dashboard extension capable of tracking detailed notebook interactions. The focus will be on the student dashboard, supporting self-reflection and awareness of one's coding

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patterns, errors, and work strategies. The overarching goal is to determine whether such dashboards can improve reflective learning, reduce repeated mistakes, and provide insights into students' perceptions of the usefulness and usability of the dashboard. To evaluate the effectiveness of this system, a controlled study will be conducted. Participants will complete two programming units in Jupyter Notebook. The first unit introduces basic Python concepts, while the second unit builds on these skills with slightly more advanced exercises. The experimental group will have access to the student dashboard for reflection during the learning units, and after each session, they will receive a more detailed view, while the control group will complete the tasks without feedback. Both quantitative measures, such as error reduction, execution efficiency, and task correctness, and qualitative measures, including self-reported reflection and usability feedback via a survey will be collected.

2 LITERATURE REVIEW

Computational notebooks have gained significant relevance in education and research, and recent work has explored how learners interact within these environments. A growing body of research shows that while notebooks support exploratory coding and flexible workflows, they also introduce challenges for understanding the learning process at a fine-grained level. Systems designed to capture cell-level interaction data—such as execution patterns, time on task, and navigation behavior—demonstrate that detailed traces can reveal iterative retries, hesitation, and fragmented workflows that remain invisible in final notebook submissions [1]. However, existing approaches of this type are primarily designed for instructor-facing analytics and do not address how such data could support student reflection or metacognitive skill development.

In programming education more broadly, research consistently shows that novice programmers struggle with error identification and debugging, often relying on unsystematic trial-and-error strategies and showing difficulty in locating the source of errors [6]. Additional studies highlight that notebook environments exhibit recurring categories of bugs, suggesting that the structure of notebooks can reinforce certain misunderstandings or error patterns [2]. These findings underline the importance of tools capable of making the coding process more transparent and helping learners understand not only the errors they make, but also how they arrive at those errors.

Learning analytics dashboards have been widely explored as mechanisms for supporting student awareness, self-regulation, and instructional decision-making. Systematic reviews show that dashboards can enhance metacognitive insight and provide users with actionable feedback on their learning behavior [8]. Despite their potential, most dashboards rely on high-level LMS metrics—such as resource access, assignment completion, or overall activity patterns—which limits their capacity to capture the step-by-step nature of programming work [4]. As a result, these systems offer only coarse-grained representations of learning progress.

Research targeting programming courses shows similar limitations. Studies that use notebook-based assignments for analytics tend to track performance at the assignment or submission level, without capturing the underlying cell-level workflow [9]. Other dashboard systems designed to support programming instructors focus on global participation and performance distributions but do not visualize detailed notebook interactions or support student-directed reflection [3]. Together, these approaches illustrate the current focus on high-level, instructor-centered metrics and the scarcity of fine-grained, student-centered analytics for notebook environments.

Across all these lines of research, a consistent gap emerges: existing dashboards rarely provide detailed visualizations of how students work within computational notebooks, nor do they investigate whether such visualizations can improve student reflection, error awareness, or learning outcomes. The predominant emphasis remains on monitoring final

outputs, aggregated performance, or instructor dashboards, with limited attention to visualization tools that help learners themselves understand their coding behaviors.

This gap motivates the present project. By developing a student-facing Jupyter dashboard extension that visualizes cell-level interaction histories, execution errors, retry patterns, and timelines of activity, the proposed study aims to contribute empirical evidence on how fine-grained notebook analytics can support metacognitive reflection and reduce repeated programming mistakes. Unlike existing systems, the focus is not solely on teacher monitoring, but on enabling learners to interpret their own behavior, gain awareness of unproductive patterns, and better understand their problem-solving process. The project also complements prior research by evaluating the dashboard through a controlled study, providing insights into both behavioral effects and students' subjective perceptions of usefulness and usability.

3 METHODOLOGY

To investigate the effectiveness of the proposed Jupyter dashboard extension in supporting student reflection and reducing programming errors, a mixed-methods research design will be employed. Our study will mainly focus on answering the following research questions:

- RQ1: Does the inclusion of a learner-facing dashboard in a Jupyter-based programming environment help students better understand and reflect on their own learning process?
- RQ2: How do students perceive the usefulness and usability of the dashboard for identifying and correcting mistakes?
- RQ3: Which usage indicators can serve as signals for frustration, reflection or effective learning strategies?

In order to address these questions, our study will mainly consist of two phases: a first phase focused on starting using the dashboard in real classroom settings, and a second phase involving a more advanced notebook whose aim is the reflection of the students' learning process. In order to evaluate the impact of the dashboard, we will divide our participants into two main groups:

- Control Group: Participants in this group will use Jupyter Notebooks without access to the dashboard features. They will complete programming exercises as usual, without any additional feedback or visualizations.
- Experimental Group: Participants in this group will have access to the Jupyter dashboard extension while completing the same programming exercises. They will be encouraged to use the dashboard to monitor their activity, reflect on their errors, and adjust their strategies accordingly.

It is important to note that both groups will work on the same programming tasks and exercises, and that the group division will be randomized to minimize selection bias, ensuring that any differences in outcomes can be attributed to the presence or absence of the dashboard. Our study design is divided in a defined and structured number of steps, which are described in the following sections.

3.1 Initial Survey

Before starting the learning units, all participants will complete a pre-survey to collect prior programming experience and baseline attitudes towards programming and self-assessment. This survey will help us control for potential confounding variables and understand the initial context of our participants. This information will allow us to analyze whether the dashboard's effectiveness varies based on students' initial experience levels or attitudes. Regarding instructors, we will also conduct a brief survey to gather how they usually grade student's notebooks, what difficulties they face when assessing student work, and what feedback (if any) they provide to students about their coding process.

This first survey will set the stage for our subsequent interventions and analyses. It will provide valuable context for interpreting the results of the learning units and will also help us tailor the dashboard features to better meet the needs of our participants.

It should be highlighted that no demographic data will be collected in order to preserve the anonymity of the participants, so no ethic approval will be needed for this study.

3.2 First Learning Unit

The first learning unit will introduce participants to basic Python programming concepts, such as variables, data types, and simple expressions. Participants will complete a series of programming exercises within Jupyter Notebooks. As mentioned earlier, the control group will not have access to the dashboard, while the experimental group will be able to use it to monitor their activity. Depending of the instructors' preferences, the experimental group students will also be given some feedback according to the information shown in the professor's dashboard. This feedback may include hints about common errors, suggestions for improvement, or prompts to reflect on their coding strategies.

The content of both dashboards will be designed depending on the results of the initial survey, but it may include metrics such as the total number of executed cells, error rates or even a timeline of activity.

3.3 Second Learning Unit

The second learning unit will build upon the concepts introduced in the first unit, introducing slightly more advanced topics such as loops and conditionals. Participants will again complete programming exercises within Jupyter Notebooks. The experimental group will continue to have access to the dashboard, allowing them to review their previous performance and errors from the first unit. This will enable us to examine whether learners who could review their behavior and performance in the first unit make fewer errors and perform better in the second unit.

3.4 Post-Survey

After completing both learning units, all participants will complete a post-survey to assess their experiences with the dashboard (for the experimental group) and their overall perceptions of the learning process. The survey will include Likert-scale items and open-ended questions to gather both quantitative and qualitative data, and will focus on aspects such as the perceived usefulness of the dashboard, its impact on self-reflection, and any changes in learning strategies.

After collecting and analyzing the data from both learning units and the surveys, we will be able to draw conclusions about the effectiveness of the Jupyter dashboard extension in supporting student reflection and reducing programming errors. The findings will inform future iterations of the dashboard and contribute to our understanding of how technology can enhance programming education.

4 CONCLUSION

This research proposal outlines a structured and empirically grounded plan to investigate how a learner-facing dashboard integrated into JupyterLab can support students' reflective practices and improve their programming processes. By combining fine-grained notebook interaction logs with both quantitative and qualitative measures of learning behavior, the study aims to provide novel evidence on how students engage with computational notebook environments beyond their final code outputs. The controlled experiment comparing students who use the dashboard with those who do not seeks to determine whether visualizing one's coding patterns and errors can foster metacognitive awareness, reduce repeated mistakes, and enhance learning strategies.

The expected outcomes of this work will contribute to the growing body of research on learning analytics in programming education by shifting the focus from course-level metrics to in-situ, process-oriented indicators. Additionally, the findings have the potential to inform the design of future dashboard systems and pedagogical interventions that make students' invisible learning processes more transparent to both learners and instructors. Ultimately, this project aims to advance our understanding of how reflective tools integrated directly within computational notebooks can support deeper learning and more effective problem-solving in introductory programming contexts.

Building on the proposed methodology, the next steps involve developing and carrying out the first initial survey to collect information about the participants' prior experiences and both their and their instructors' needs regarding the dashboard features. This information will then guide the design of the dashboard's metrics and visualizations to ensure they are relevant and actionable for the target users. Following this, and after the implementation of the dashboard extension and the recruitment of participants, the two learning units and associated data collection activities can be conducted.

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